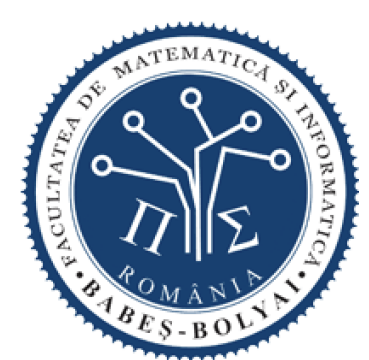




Operations Research

01 - Introduction to Operations Research

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Structure

- Courses 2 h/week
- Seminars 1 h/week
- course web page: <https://www.cs.ubbcluj.ro/~mihai-suciu>
- materials will be posted on MsTeams (access code 8pld2cd)
- activities will take place according to
<https://www.cs.ubbcluj.ro/files/orar/2025-1/disc/MME8215.html>

Evaluation

- seminar:
 - projects (implement/solve optimization problems using the learned models)
- bonus for extra work (0,5p)
- written exam
- final grade:

$$\text{Grade} = 0.6 * E + 0.4 * S + B$$

where:

E = written exam

S = seminar

B = bonus

Minimum standard of performance

$$\text{Grade} = 0.6 * E + 0.4 * S + B$$

- Grade: minimum 5 at each grading activity (except B).

Course content

Tentative schedule, courses

1. Introduction to Operations Research
2. Introduction to Foundation mathematics and statistics
3. Linear Programming
4. Maximization Then Minimization problems.
5. Graphical LP Minimization solution, Introduction, Simplex method definition, formulating the Simplex model
6. Linear Programming – Simplex Method for Maximizing.
7. Transportation model
8. Network model
9. Advanced linear programming, Integer linear programming
10. Metaheuristics Optimization Algorithms
11. Multi-objective optimization
12. Optimization with engineering Applications, as for example Routing and traffic management, Network design, Network connectivity and reliability, Energy consumption consideration in network optimization.
13. Use cases extracted from research papers.
14. Use cases extracted from research papers.

Course content

Tentative schedule, seminar

1. Problem definition, unconstrained optimization, constrained optimization
2. Modeling combinatorial optimization problems
3. Linear programming, Integer and mixed-integer linear programming.
4. Optimization with engineering applications
5. Metaheuristics
6. Multiobjective optimization
7. Different use cases

Let's begin

- 01 Introduction to Operations Research
 - What is OR?

Intro to OR

- The OR technique was initiated in England during II world war to solve strategic and tactical military problems.
- Then British and American military also started the same technique for effective utilization of military resources.
- Specialists in engineering, mathematics, economics, psychology, statistics and physical science were formed as group to do research on military problems.
- The same technique was spread to United states, France and Canada.
- After the world war these teams were shifted towards industry and business sector to solve complex managerial problems.

Intro to OR

- colleges and universities introduced OR as a subject at degree level (~1950)
- OR club was formed in England, later its name was changed as OR society of U.K (~1948)
- International federation of OR society was established with headquarters in London (~1957)
- Today OR techniques are being used in various fields such as hospitals, libraries, city planning, transportation systems, crime investigation and industries.

Definitions

- “OR utilizes the planned approach and an interdisciplinary team in order to represent complex functional relationships as mathematical models for the purpose of providing a quantitative analysis”. - Thierauf and Klekamp
- “OR is a scientific approach to problem solving for executive management”. - H.M.Wagner
- “OR is a scientific knowledge through interdisciplinary team effort for the purpose of determining the best utilization of limited resources. - H.A.Taha
- “OR is the art of winning wars without actually fighting them.” - Auther Clark
- “OR is the art of finding bad answers to problems to which otherwise worse answers are given.” - T.L.Saaty
- “OR is essentially a collection of mathematical techniques and tools which in conjunction with system approach, are applied to solve practical decision problems of an economic or engineering nature”. - Daellenbach and George

What is OR

- In general, the organization goal is to optimize the use of available resources.
- OR (Management science) is a discipline that adapts the scientific approach to problem solving to decision making.

What is OR

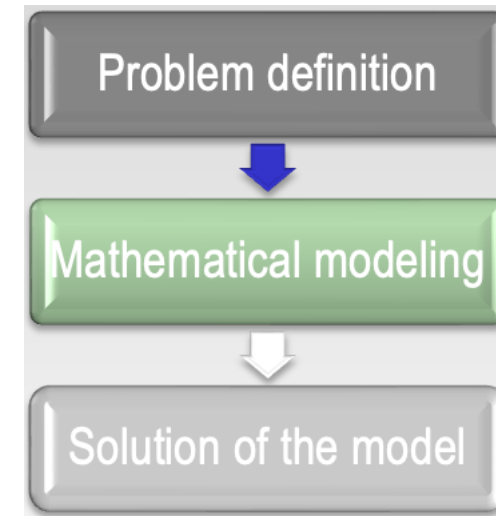
- Application of quantitative mathematical methods and logic to find solutions to managerial problems
- Concerned with means of dealing with:
 - Resource constraints
 - Assignment and maximization of values
 - Appraisal of alternatives in an uncertain, dynamic environment

Benefits of OR

- Ability to deal in a quantitative manner with conditions of uncertainty
- Use of objective methods to sort out in complex situations
- Capacity to illustrate the likely outcomes of alternative courses
- Assist managers in understanding many inter-related factors
- Provision of various logical approaches to make decisions
- Ability to handle masses of data with help of computer

Methodology

1. Formulate the problem.
2. Construct the model.
3. Acquire input data.
4. Solve the model.
5. Test the solution.
6. Implement the solution.
7. Modifying the model.
8. Establishing control over the solution.



1. Defining the problem

- OR is Applied When:
 - Designing and implementing new operations.
 - Evaluating ongoing Operations and Procedures.
 - Determining and recommending corrective actions

1. Defining the problem

- How to Start and How to Proceed
 - Identify the problem.
 - Observe the problem from various points of view.
 - Keep things simple.
 - Identify constraints.
 - Work with management, get feedback

2. Building a Mathematical Model

- Identify Decision Variables
 - Which factors are controllable?
- Quantify the Objective and Constraints
 - Formulate the function to be optimized (profit, cost).
 - Formulate the requirements and/or restrictions.
- Construct a Model Shell
 - Help focus on the exact data required.
- Gather Data
 - Consider time / cost issues.

2. Building a Mathematical Model

- Mathematical modeling formulation: The general OR model can be organized in the following general format:-

Maximize or minimize *Objective Function*

subject to

Constraints

3. Solving a Mathematical Model

- Choose an Appropriate Solution Technique
 - An optimization algorithm.
 - A heuristic algorithm.
- Generate Model Solutions
- Test / Validate Model Results
 - Is the solution reasonable?
 - Are radical changes needed?
 - Does it fit present and future plans?
 - Unacceptable results? Return to modeling.
- Perform “What--If” Analyses

Examples

1. Wilson company produces three products: desk, chairs, and molded steel.

It is trying to decide on the number of desks (D), chair (C), and pounds of molded steel (M).

Assumption :

1- If the nets \$ 50 profit on each desk produced, \$30 on each chair produced, and \$ 6 per pound of modeled steel produced. **What the total profit?**

The answer :

- Total profit for production run can be modeled by the expression

$$50 D + 30 C + 6 M$$

2 - if 7 pounds of raw steel are needed to manufacture a desk, 3 pounds to manufacture a chair, and 1.15 pounds to produce a pound of model steel.

What the amount of raw steel used during the production?

The answer:

- The amount of raw steel used during the production run is modeled by the expression:

$$7D + 3C + 1.5 M$$

3 - Wilson has only 2000 pounds of raw steel available. And at least 100 desk must be produced to satisfy contract commitment.

The answer:

- The function constraint can be modeled as :

$$7 D + 3C + 1.5 M \leq 2000 \text{ (raw steel)}$$

$$D \geq 100 \text{ (contract)}$$

The **Mathematical model** for this problem is formulated as :

MAXIMIZE

$$50 D + 30 C + 6 M$$

SUBJECT TO

$$7D + 3C + 1.5M \leq 2000 \text{ (Raw steel)}$$

$$D \geq 100 \text{ (contract)}$$

Example

2. Consider the following **tickets purchasing problem**.

A businessperson has a 5-week commitment traveling between Fayetteville (FYV) and Denver (DEN). Weekly departure from Fayetteville occurs on Mondays for return on Wednesdays. A regular roundtrip ticket costs \$400, but a 20% discount is granted if the roundtrip dates span a weekend. A one-way ticket in either direction costs 75% of the regular price.

How should the tickets be bought for the 5-week period?

Decision making problem, the solution requires answering three questions:

1. What are the decision **alternatives**?
2. Under what **restrictions** is the decision made?
3. What is an appropriate **objective criterion** for evaluating the alternatives?

- Three **plausible alternatives**:
 1. Buy five regular FYV-DEN-FYV for departure on Monday and return on Wednesday of the same week.
 2. Buy one FYV-DEN, four DEN-FYV-DEN that span weekends, and one DEN-FYV.
 3. Buy one FYV-DEN-FYV to cover Monday of the first week and Wednesday of the last week and four DEN-FYV-DEN to cover the remaining legs. All tickets in this alternative span at least one weekend.

The **restriction** on these options is that the businessperson should be able to leave FYV on Monday and return on Wednesday of the same week.

- An obvious **objective criterion** for evaluating the proposed alternatives is the **price** of the tickets.
- The alternative that yields the smallest cost is the best.
- We have:
 - Alternative 1 cost= $5 * \$400 = \2000
 - Alternative 2 cost= $0.75 * \$400 + 4 * (0.8 * \$400) + 0.75 * \$400 = \1880
 - Alternative 3 cost= $5 * (0.8 * \$400) = \1600

Alternative 3 is the cheapest

- Based on the preceding two examples, the general OR model can be organized in the following general format:

Maximize / minimize

Objective Function

subject to

Constraints

A solution is **feasible** if it satisfies all the constraints. It is **optimal** if, in addition to being feasible, it yields the best (maximum or minimum) value of the objective function.

- In the ticket purchasing problem, the problem considers three feasible alternatives, with the third alternative being optimal.
- Though OR models are designed to optimize a specific objective criterion subject to a set of constraints, the quality of the resulting solution depends on the degree of completeness of the model in representing the real system.
- In the ticket purchasing problem: if all the dominant alternatives for purchasing the tickets are not identified, then the resulting solution is optimum only relative to the alternatives represented in the model, e.g. leave out option 3 -> **suboptimal** solution.

Methods in OR

- Statistical techniques
 - Time trends and forecasting
 - Decision theory and tree
- Activity analysis
 - Time motion studies
 - Work sampling and activity analysis
- Mathematical techniques
 - Simulation study/models
 - System analysis
 - Linear programming
 - Inventory control
 - Network analysis

Methods in OR

- Financial techniques
 - Monitoring expenditure
 - Cost accounting and analysis
 - DALY
 - Cost benefit analysis
 - Cost effectiveness analysis
 - Zero base budgeting
 - Input output analysis
 - Outcome budget
- Miscellaneous
 - Management by exception
 - Situational analysis
 - Current state assessment
 - SWOT analysis
 - Log frame analysis

Some other techniques used in O.R

1. Linear programming.
2. Dynamic programming.
3. Queuing theory.
4. Inventory theory.
5. CPM and PERT techniques.
6. Game theory.
7. Replacement models.
8. Sequencing models.
9. Decision analysis models.
10. Simulation.

Characteristics or significant features of OR

- **Decision making tool**: Optimal solution to the problem.
- **Scientific approach**: Scientific methods, techniques and tools for analyzing and solving the problems.
- **Interdisciplinary team approach**: Members of an interdisciplinary team analyze the cause-and-effect relationship between various parameters of the problem.
- **System approach or wholistic approach**: Indirect effect on all other parts is considered.
- **Objective**: Find the best or optimal solutions to the problem.

General methods for solving OR models

- **Analytical Method**: Mathematical tools like calculus, matrix algebra, probability, graphs etc. are used for solving OR problems.
- **Numerical or Iterative method**: when analytical methods fail to solve complex problems numerical methods are used. Numerical methods involve a trial and error procedure. An iterative process is one in which successive trials tend to approach an optimal solution.
- **Monte-Carlo simulation**: use of probability and random sampling concepts.

Scope

- A decision maker can conduct the experiments on the model by changing various conditions to examine the effect of these changes and determine the best solution.
- The experiments can be performed on the model, avoiding causing any damage to an existing system.
- Computer simulations can be used to solve complex problems.

Shortcomings

- Building complex OR models is economically feasible only for repetitive problems. Limitations on resources like money, time etc.
- OR models may not represent the realistic situations in which decisions have to be taken.
- The data is subject to frequent changes which makes the process of updating the models quite expensive.
- The decision maker may not be fully aware of the limitations of OR model being used.
- The inability of OR to convey and communicate results to business executives who need to make decisions.